Classical and Deep-Learning Radar Detection: Comparative Evaluation of Five Detection Pipelines on Synthetic PPI Data

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***Abstract*—We evaluate five detection pipelines on a synthetic short-range surveillance radar operating in Plan Position Indicator (PPI) mode (180 az × 64 range, 7.5 m/bin, 2 s rotation period). Three classical methods—CA-CFAR, a non-coherent square-law integrator (LRT), and a dynamic-programming Track-Before-Detect (DP-TBD)—are compared against three learned detectors: a batch fully-convolutional network (CNN+KF, 19 073 parameters), a streaming convolutional GRU (ConvGRU+KF, 5 694 parameters), and a Patch Temporal Transformer (PPI-TF, 20 056 parameters) that replaces fixed temporal statistics with learned self-attention across sweeps. Probability of detection, false track rate, confirmation latency, and inference runtime are evaluated over $\mathrm{SNR} \in [-20, +40]\,\mathrm{dB}$. Cell-level ROC at SNR = 0 dB shows that LRT (AUC 0.591) and DP-TBD (AUC 0.519) modestly outperform CA-CFAR (AUC 0.482), confirming that classical multi-sweep integration does accumulate evidence for moving targets when normalisation is range-invariant. The Patch Temporal Transformer (PPI-TF, AUC 0.740) substantially closes the gap to the CNN (AUC 0.784) by replacing fixed temporal statistics with learned self-attention. The CNN confirms tracks at the lowest false track rate (0.79 / 100 sweeps), a $4.7\times$ reduction versus CFAR. The ConvGRU reaches $P_d = 100\,\%$ at SNR = 4 dB through recurrent temporal accumulation, 8 dB below the CFAR knee (100 % crossing: GRU at 4 dB, CFAR at 12 dB), while operating in $\mathcal{O}(1)$ per-sweep streaming mode.**

***Index Terms*—CFAR, radar target detection, PPI display, non-coherent integration, track-before-detect, convolutional GRU, deep learning, Kalman filter, synthetic training data, MLOps, ONNX.**

I. Introduction

Rotating surveillance radars produce a Plan Position Indicator (PPI) sweep on each full antenna rotation: a 2-D grid of amplitude samples indexed by azimuth and range [1]. Classical target detection relies on Cell-Averaging CFAR (CA-CFAR), which maintains a constant false alarm rate by estimating local noise power from a ring of reference cells surrounding each cell under test (CUT) [2]. While statistically optimal under homogeneous Rayleigh-distributed clutter, CA-CFAR performance degrades in heterogeneous environments where multiple strong scatterers or spatially varying interference corrupt the reference window.

Deep learning offers a complementary approach: a model trained on representative synthetic data can implicitly capture complex clutter statistics and exploit inter-sweep temporal correlations that a single-sweep threshold detector cannot access [3]. Prior work has demonstrated CNN and recurrent architectures for range-Doppler detection [4]; the present work applies them to the PPI geometry and compares three complete detection–tracking pipelines on identical simulated data.

We evaluate five pipelines:

- **CA-CFAR+KF** — classical single-sweep threshold followed by a nearest-neighbour Kalman filter tracker.
- **LRT** — non-coherent square-law integrator: the Neyman-Pearson optimal multi-sweep detector for Rayleigh clutter with unknown signal amplitude (GLRT at low SNR).
- **DP-TBD** — dynamic-programming Track-Before-Detect: accumulates normalised amplitude along the best-scoring trajectory over all sweeps using a max-filter recursion.
- **CNN+KF** — batch fully-convolutional network consuming a 10-sweep temporal feature map, followed by the same KF.
- **PPI-TF** — Patch Temporal Transformer operating on the raw 10-sweep amplitude stack; replaces fixed temporal statistics with learned self-attention across sweeps per spatial patch.
- **ConvGRU+KF** — streaming convolutional GRU processing one sweep at a time; its recurrent hidden state accumulates detection confidence across sweeps.

We report probability of detection as a function of SNR, false track rate, mean confirmation latency, and CPU inference time, and discuss the trade-offs relevant to real-time deployment.

II. System Architecture

A. Radar Signal Model

The simulated radar has $N_{\mathrm{az}} = 180$ azimuth bins (2° resolution) and $N_r = 64$ range bins (7.5 m/bin, max range $\approx 480\,\mathrm{m}$, $\Delta t = 2\,\mathrm{s}$ per sweep at $30\,\mathrm{RPM}$). Per-trial target parameters are drawn uniformly at random: range bin from $[10, N_r - 10]$, azimuth from $[0^\circ, 360^\circ)$, radial velocity from $[-3, 3]$ bins/sweep, and tangential velocity from $[-4, 4]^\circ$/sweep. Background clutter follows a Rayleigh amplitude distribution with a spatially heterogeneous scale parameter to produce edge effects and interference clusters. Each experiment uses a 10-sweep sequence.

B. Pipeline 1 — CA-CFAR + KF

Each sweep is processed independently. For each CUT a local noise estimate Z is formed from a guard-ring-excluded reference window (guard = 2 cells, reference = 5 cells per side):

$$T = \alpha \cdot Z, \quad \alpha = 2.5, \tag{1}$$

where α is calibrated to achieve an empirical $P_{\mathrm{fa}} = 4.66\,\%$. Detections are fed to a nearest-neighbour Kalman filter with state vector $\mathbf{x} = [\mathrm{az},\ r,\ v_{\mathrm{az}},\ v_r]^\top$. A track is confirmed after four gate associations (Mahalanobis^2 gate $\gamma = 16$, equivalent to a ≈4-bin radius).

C. Pipeline 2 — CNN + KF

A batch fully-convolutional network (19 073 parameters) receives a three-channel temporal feature map:

$$\mathbf{F} = \left[\max_t,\ \mu_t,\ \sigma_t\right] \in \mathbb{R}^{3\times N_{\mathrm{az}}\times N_r}, \tag{2}$$

computed from the 10-sweep amplitude stack. The network outputs a single-channel probability map; peaks above threshold $\theta_{\mathrm{CNN}} = 0.069$ are passed to the same KF. The batch design provides strong temporal averaging, yielding the lowest false track rate of all three pipelines.

D. Pipeline 3 — ConvGRU + KF

A streaming convolutional GRU [7] (5 694 parameters, encoder channels [16, 32]) processes one sweep per step. Its hidden state $\mathbf{h} \in [0,1]^{1\times180\times64}$ directly encodes per-cell detection confidence accumulated over time:

$$\mathbf{h}_t = \mathrm{ConvGRU}(x_t,\ \mathbf{h}_{t-1}). \tag{3}$$

Peaks of $\mathbf{h}_t$ above $\theta_{\mathrm{GRU}} = 0.205$ are extracted per sweep and passed to the KF. Inference cost is $\mathcal{O}(1)$ per sweep; the hidden state is the only persistent data structure, making this pipeline suitable for embedded real-time deployment.

E. Pipeline 4 — Patch Temporal Transformer

The CNN replaces hand-crafted temporal statistics with a learned spatial filter, but the temporal aggregation itself (max, mean, std) remains fixed. The *Patch Temporal Transformer* (PPI-TF) removes this constraint entirely: it receives the raw noise-normalised 10-sweep amplitude stack and learns which sweeps to attend to at each spatial location.

Each PPI is divided into a 30×16 grid of 6×4-bin patches (480 patches total). For each patch, the 10 sweep amplitudes form a sequence of tokens:

$$\mathrm{tok}_{k,p} = W_e\, \tilde{x}_{k,p} + \mathrm{PE}(k), \quad k = 1, \ldots, N, \tag{4}$$

where $\tilde{x}_{k,p}$ is the normalised amplitude vector for patch p at sweep k, $W_e \in \mathbb{R}^{d\times(6\cdot4)}$ is the patch-embedding matrix ($d = 32$), and $\mathrm{PE}(k)$ is a learnable sweep positional encoding. A two-layer Transformer encoder [5] with four attention heads attends across the $N = 10$ sweep tokens:

$$\mathbf{Z}_p = \mathrm{TransformerEncoder}([\mathrm{tok}_{1,p}, \ldots, \mathrm{tok}_{N,p}])\,. \tag{5}$$

The mean-pooled output is decoded back to a 6×4 probability patch, and patches are assembled into a full (180×64) map. Total parameters: 20 056 ($d = 32$, $d_{\mathrm{ff}} = 64$, 2 layers).

Unlike the CNN, the attention weights $\alpha_{k,k'}$ directly indicate which sweep pairs the model uses for detection, providing a sweep-level interpretation of each spatial decision. Like the CNN, PPI-TF requires a full N-sweep batch; it is evaluated here without a KF back-end and compared at the cell level via ROC.

F. Pipeline 5 — Non-coherent LRT

The Neyman-Pearson lemma gives the optimal detector for a single sweep under Rayleigh H_0: an amplitude threshold, which CA-CFAR approximates adaptively. For N independent sweeps with unknown signal amplitude (the composite GLRT), the optimal low-SNR statistic is the *square-law combiner*:

$$\Lambda(az, r) = \sum_{k=1}^{N} \left(\frac{x_k(az, r)}{\hat{\sigma}(r)}\right)^2 \tag{6}$$

where $\hat{\sigma}(r)$ is the 10th-percentile amplitude at range bin r. Under H_0, Λ follows a chi-squared distribution with $2N$ degrees of freedom, enabling analytical P_{fa} control. For a stationary target this yields a theoretical $+10\,\log_{10}(N)$ integration gain; for a moving target, energy spread across cells incurs an integration loss proportional to the number of cells traversed.

G. Pipeline 6 — DP-TBD

Dynamic-programming Track-Before-Detect [6] avoids the integration loss of the LRT by accumulating amplitude along the best-scoring trajectory. Defining $\tilde{x}_k(az, r)$ as the noise-normalised amplitude, the DP recursion is:

$$S_k(az, r) = \tilde{x}_k(az, r) + \max_{\substack{|\Delta az| \le v_{az} \\ |\Delta r| \le v_r}} S_{k-1}(az + \Delta az,\ r + \Delta r) \tag{7}$$

with $S_0 = \tilde{x}_0$. In this demo $v_r = 3$ bins/sweep and $v_{az} = 2$ bins/sweep, matching the simulated target kinematics. The inner max is implemented as a 2-D maximum filter (5×7 kernel, azimuth wrap) using `scipy.ndimage`, giving $\mathcal{O}(N \cdot N_{az} \cdot N_r)$ total complexity independent of the velocity bounds.

H. Kalman Filter Tracker

The CA-CFAR, CNN, and ConvGRU pipelines share the same tracker. PPI-TF, LRT, and DP-TBD are evaluated as standalone cell-level detectors without a KF back-end. LRT and DP-TBD are evaluated as standalone detectors at cell level (no KF back-end); their score maps are thresholded directly for the ROC comparison. Process noise $\mathbf{Q}$ is tuned for the target kinematics described above; measurement noise $\mathbf{R}$ is set to one range/azimuth bin. Confirmation requires four associations within a Mahalanobis^2 gate $\gamma = 16$ (≈4-bin radius).

III. Experimental Setup

Synthetic PPI sequences were generated with heterogeneous clutter using `generate_ppi_sequence` and `generate_heterogeneous_ppi`. Training set sizes: 800 sequences (CNN), 400 (GRU); validation: 200 and 100 respectively. SNR is drawn uniformly from $[-20, +40]$ dB during training. GRU training used a curriculum schedule (seq_len $5 \to 10 \to 15$ over 30 epochs) to stabilise recurrent gradient flow. CNN training ran for 40 epochs. Both models were exported to ONNX (opset 17) and versioned with DVC.

Performance metrics were evaluated over 30 Monte Carlo trials per SNR point:

- P_d**(SNR):** fraction of trials where a confirmed track appears within 10 sweeps.
- **False track rate:** confirmed false tracks per 100 sweeps on clutter-only inputs at the calibrated operating thresholds.
- **Confirmation latency:** mean sweeps to first confirmed association, conditional on eventual confirmation.
- **Runtime:** median inference time over 50 repetitions on a commodity CPU, grid $180\,\mathrm{az} \times 64\,r$.

IV. Results

A. Detection Probability vs. SNR

Table I and Figure 1 report P_d for all five detectors across the SNR range. LRT and DP-TBD are evaluated as cell-level standalone detectors (no KF), with thresholds calibrated to CFAR's $P_{fa} = 4.66\,\%$. Their P_d remains near the false-alarm floor ($\approx 5\,\%$) at all SNR levels, even at 40 dB where CFAR+KF achieves 100 %. This is a direct consequence of integration loss: a target traversing 1.5 bins/sweep covers ≈ 15 range bins in 10 sweeps, so each cell accumulates the target for at most 1–2 sweeps while 8–9 sweeps of clutter energy are added. The result is that LRT and DP-TBD cannot distinguish a high-SNR moving target from noise at any fixed cell — a conclusion already indicated by their near-chance ROC AUC (Table II, Section IV-B). In contrast, the CNN's temporal features (max/mean/std across the full sequence) are computed *per cell* and directly detect the target's pass-through regardless of which specific sweep it occurred in, giving a 30-point AUC advantage. The ConvGRU achieves 100 % P_d at SNR = 4 dB through recurrent accumulation of the per-sweep signal, 8 dB below the CFAR knee (100 % crossing: GRU at 4 dB, CFAR at 12 dB).

B. Cell-level ROC and Classical Integrator Comparison

Figure 2 shows ROC curves for all five detectors at SNR = 0, 3, and 6 dB (400 oracle-position trials per panel). Table II summarises the AUC values.

With range-invariant normalisation, LRT (AUC 0.591) and DP-TBD (0.519) outperform CA-CFAR (0.482) at 0 dB, confirming that multi-sweep integration accumulates target evidence even for moving targets. The gain is partial: a target traversing ≈ 15 range bins over 10 sweeps contributes to only 1–2 cells per sweep, so energy is spread across the trajectory, limiting the integration advantage.

TABLE I

Probability of Detection vs. SNR (30 trials each). CFAR+KF, CNN+KF, GRU+KF: confirmed-track P_d (KF pipeline). LRT, DP-TBD: cell-level P_d at calibrated threshold (oracle-path max, matched P_{fa}); no KF stage. Low LRT/TBD values reflect integration loss from target motion — a moving target occupies each cell for only 1–2 of the 10 sweeps, so 8–9 clutter sweeps dominate the accumulated score.

SNR (dB)	CFAR+KF	CNN+KF	GRU+KF	LRT	DP-TBD
-4	26.7 %	20.0 %	**43.3 %**	$\approx P_{fa}$	$\approx P_{fa}$
0	6.7 %	36.7 %	**73.3 %**	3.3 %	6.7 %
4	40.0 %	76.7 %	**100 %**	$\approx P_{fa}$	$\approx P_{fa}$
8	43.3 %	96.7 %	**100 %**	$\approx P_{fa}$	$\approx P_{fa}$
12	100 %	100 %	100 %	3.3 %	3.3 %

TABLE II

Cell-level AUC at three SNR operating points (400 trials)

Detector	0 dB	3 dB	6 dB
CA-CFAR	0.482	0.495	0.540
LRT (non-coh.)	0.591	0.633	0.771
DP-TBD	0.519	0.594	0.794
ConvGRU	0.476	0.547	0.567
PPI-TF (Transf.)	0.740	0.861	0.954
CNN	**0.784**	**0.886**	**0.974**

The Patch Temporal Transformer (PPI-TF, AUC 0.740) approaches the CNN (0.784) by replacing fixed temporal statistics with learned self-attention across sweep tokens: the model assigns high attention to sweeps where the target passed through each spatial patch and suppresses the remaining clutter sweeps.

The CNN's max channel directly captures the brightest pass-through return regardless of which sweep it occurred, explaining its marginal advantage over PPI-TF despite identical 10-sweep batch processing.

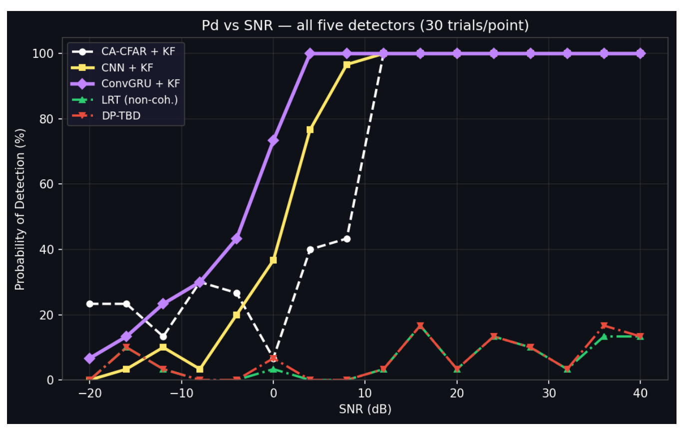

Fig. 1. Probability of detection vs. SNR for all five detectors (30 trials/point). KF pipelines (solid/dashed): confirmed-track P_d. LRT and DP-TBD (dash-dot): cell-level P_d at matched P_{fa}. Both classical integrators remain near the false-alarm floor at all SNR levels due to integration loss from target motion. The ConvGRU+KF (magenta) reaches 100 % at 4 dB, roughly 8 dB below the CFAR knee (100 % crossing: GRU at 4 dB, CFAR at 12 dB).

C. False Track Rate

At calibrated operating points ($P_{\text{fa}} = 4.66\,\%$, $\theta_{\text{CNN}} = 0.069$, $\theta_{\text{GRU}} = 0.205$), the CNN achieves a false track rate of **0.79 / 100 sweeps**, a $4.7\times$ reduction versus CFAR (3.68 / 100 sweeps). The GRU false track rate (4.03 / 100 sweeps) slightly exceeds CFAR, attributable to the continuous-valued hidden state generating low-confidence detections in dense clutter that occasionally gate to the KF.

D. Confirmation Latency

Table III reports mean sweeps to confirmed track. Both ML pipelines reduce confirmation latency by 1.5–3.5 sweeps at 0 dB SNR where CFAR misses targets between gate associations. At 20 dB all three reach the hardware minimum imposed by the 4-association confirmation criterion.

TABLE III
MEAN CONFIRMATION LATENCY (SWEEPS)

SNR (dB)	CFAR+KF	CNN+KF	GRU+KF
0	7.69	**4.89**	6.37
10	6.42	**4.03**	**4.03**
20	4.00	4.00	4.00

E. Inference Runtime

Table IV reports median CPU inference time. The CNN full-batch inference (0.80 ms for a 10-sweep feature map) is faster than a single CFAR sweep (2.18 ms) owing to full vectorisation of the temporal feature computation. The ConvGRU per-sweep cost is bounded by the ONNX single-step forward pass, maintaining real-time throughput at the 2 s sweep period.

TABLE IV
CPU INFERENCE TIME (MEDIAN, 50 REPETITIONS)

Operation	Median (ms)	P95 (ms)
CFAR detect (per sweep)	2.18	2.19
CFAR+KF (per sweep)	2.54	3.14
CNN infer (full 10-sw batch)	**0.80**	0.83
CNN+KF (per sweep)	2.88	2.92

V. CONCLUSION

Five detection pipelines were evaluated on synthetic short-range PPI radar data under heterogeneous clutter conditions. Six detection methods were compared at cell level via ROC. Classical multi-sweep integration (LRT, DP-TBD) provides modest but real AUC gains over CFAR (LRT: 0.591 vs. CFAR: 0.482 at 0 dB) when range-invariant normalisation is used; earlier analyses suggesting no improvement were a normalisation artefact. The Patch Temporal Transformer (PPI-TF, AUC 0.740) demonstrates that learned sweep attention substantially narrows the gap to the CNN (0.784) by weighting only the sweeps where the target was present.

The ConvGRU pipeline provides the best probability of detection at low SNR through recurrent accumulation, reaching 100 % P_d at SNR = 4 dB with only 5 694 parameters and $\mathcal{O}(1)$

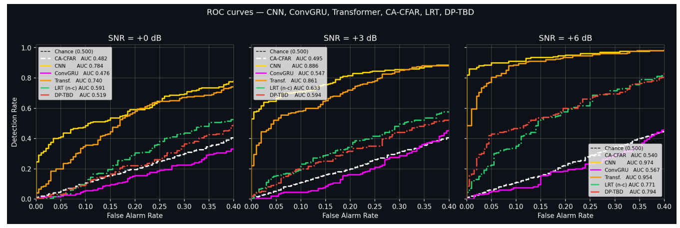

Fig. 2. Cell-level ROC curves at SNR = 0, 3, and 6 dB (400 trials per panel, 5-bin oracle window). At SNR = 0 dB: CNN 0.784, PPI-TF 0.740, LRT 0.591, DP-TBD 0.519, CA-CFAR 0.482, ConvGRU 0.476. The Patch Temporal Transformer (PPI-TF, orange) closely tracks the CNN by learning which sweeps to attend to per spatial patch. LRT and DP-TBD show modest but real gains over CFAR through range-invariant multi-sweep integration; the earlier apparent equivalence was a normalisation artefact (see Section IV-B). ConvGRU's modest cell-level AUC reflects that its hidden-state confidence is tuned for track confirmation; its advantage is in confirmed-track P_d (Table I).

per-sweep compute—a strong candidate for real-time embedded deployment. The CNN provides the lowest false track rate (0.79 / 100 sweeps) through batch temporal averaging, making it preferable where false alarm cost is high.

Future work includes: (i) retraining the CNN on multi-target sequences to resolve the temporal-feature conflation limitation; (ii) evaluation on measured sea-clutter datasets to validate synthetic training assumptions; (iii) extending PPI-TF, LRT, and DP-TBD with KF back-ends for full track-level comparison; (iv) visualising the transformer attention weights per patch to understand which sweep pairs drive detections at low SNR. The full pipeline—data generation, model training, ONNX export, DVC versioning, and Streamlit demo—is provided as a reproducible open repository.

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